

Duality

Lecture 4

October 1, 2026

Motivation

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4. *You are offered a bit more of b_i , for a “suitable price”. Is the deal worthwhile?*

Duality theory will provide answers to these questions (and more)

Outline

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$$c^T x^* = \tilde{r}^T y^* \quad \textbf{(strong duality)}$$

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- In the process, will uncover some **fundamental ideas in optimization**:

separation of convex sets $\implies$ Farkas Lemma $\implies$ strong duality

Deriving Lower Bounds

Consider a linear optimization problem in the most general form possible:

Primal Problem

$$\begin{array}{ll} (\mathcal{P}) \text{ minimize}_x & c^T x \\ \text{such that} & a_i^T x \geq b_i, \quad \forall i \in I_{ge}, \\ & a_i^T x \leq b_i, \quad \forall i \in I_{le}, \\ & a_i^T x = b_i, \quad \forall i \in I_{eq}, \\ & x_j \geq 0, \quad \forall j \in J_p, \\ & x_j \leq 0, \quad \forall j \in J_n, \\ & x_j \text{ free}, \quad \forall j \in J_f \\ \text{variable} & x \in \mathbb{R}^n. \end{array} \tag{1}$$

Note the mnemonic encoding...

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Definition

We will refer to this as the **primal problem** or problem $(\mathcal{P})$.

Let P denote its feasible set (a polyhedron), and p^* denote its optimal value.

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$(\mathcal{P})$ is a minimization; we seek **valid lower bounds** on $(\mathcal{P})$. *Any ideas?*

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For every constraint i , have a **penalty** λ_i

Construct the **lower bound** as the **Lagrangian**:

$$\mathcal{L}(x, \lambda) = c^T x - \sum_{i=1}^m \lambda_i (a_i^T x - b_i) = c^T x - \lambda^T (Ax - b)$$

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Note: we relaxed the complicating constraints, $a_i^T x \text{ (?) } b_i$, and used a linear penalty

Not apriori clear that this will give us very good bounds...

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We want the Lagrangean to give us **a valid lower bound**:

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Deriving Lower Bounds

Summarizing... any $\lambda \in \Lambda$ produces a **valid lower bound**:

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*How can we get a lower bound on the primal's **optimal value** p^* ?*

Deriving Lower Bounds

Summarizing... any $\lambda \in \Lambda$ produces a **valid lower bound**:

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Claim

The function $g : \Lambda \rightarrow \mathbb{R}$ defined as:

$$g(\lambda) := \min_x \mathcal{L}(x, \lambda) \tag{3}$$

$$s.t. \ x_j \geq 0, \ \forall j \in J_p$$

$$x_j \leq 0, \ \forall j \in J_n$$

$$x_j \text{ free}, \ \forall j \in J_f$$

satisfies $g(\lambda) \leq p^$ for any $\lambda \in \Lambda$.*

Note: including the sign constraints on x in this optimization improves the lower bound!

Deriving Lower Bounds

Let us analyze this further:

$$\begin{aligned} g(\lambda) = \min_x \mathcal{L}(x, \lambda) &= \min_x [\lambda^T b + (c^T - \lambda^T A)x] \\ \text{s.t. } x_j &\geq 0, \forall j \in J_p, & \text{s.t. } x_j &\geq 0, \forall j \in J_p, \\ x_j &\leq 0, \forall j \in J_n, & x_j &\leq 0, \forall j \in J_n, \\ x_j &\text{ free}, \forall j \in J_f & x_j &\text{ free}, \forall j \in J_f \end{aligned}$$

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$$g(\lambda) = \begin{cases} \lambda^\top b, & \text{if } \lambda^\top A_j \leq c_j, \forall j \in J_p \text{ and } \lambda^\top A_j \geq c_j, \forall j \in J_n \text{ and } \lambda^\top A_j = c_j, \forall j \in J_f \\ -\infty, & \text{otherwise.} \end{cases}$$

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is a **valid lower bound** on the primal **optimal value**: $g(\lambda) \leq p^*$ for any $\lambda \in \Lambda$.

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$$\underset{\lambda \in \Lambda}{\text{maximize}} \ g(\lambda) \tag{4}$$

This is equivalent to the following optimization problem:

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This is equivalent to the following optimization problem:

Dual Problem

$$\begin{aligned} & \text{maximize} && \lambda^T b \\ & \text{subject to} && \lambda_i \geq 0, && \forall i \in I_{ge}, \\ & && \lambda_i \leq 0, && \forall i \in I_{le}, \\ & && \lambda_i \text{ free}, && \forall i \in I_{eq}, \\ & && \lambda^T A_j \leq c_j, && \forall j \in J_p, \\ & && \lambda^T A_j \geq c_j, && \forall j \in J_n, \\ & && \lambda^T A_j = c_j, && \forall j \in J_f. \end{aligned} \tag{5}$$

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Definition

This is the **dual** of $(\mathcal{P})$, which we will also refer to as $(\mathcal{D})$. We denote its feasible set with D and its optimal value with d^* .

Note: The dual is also a linear optimization problem!

Primal-Dual Pair

Primal-Dual Pair of Problems

$$\begin{array}{ll}\text{Primal } (\mathcal{P}) \\ \text{minimize}_{\mathbf{x}} & \mathbf{c}^T \mathbf{x} \\ (\lambda_i \rightarrow) & \mathbf{a}_i^T \mathbf{x} \geq b_i, \quad \forall i \in I_{ge} \\ (\lambda_i \rightarrow) & \mathbf{a}_i^T \mathbf{x} \leq b_i, \quad \forall i \in I_{le} \\ (\lambda_i \rightarrow) & \mathbf{a}_i^T \mathbf{x} = b_i, \quad \forall i \in I_{eq} \\ & x_j \geq 0, \quad \forall j \in J_p \\ & x_j \leq 0, \quad \forall j \in J_n \\ & x_j \text{ free}, \quad \forall j \in J_f \\ \text{variables} & \mathbf{x} \in \mathbb{R}^n\end{array}$$

$$\begin{array}{ll}\text{Dual } (\mathcal{D}) \\ \text{maximize}_{\lambda} & \lambda^T \mathbf{b} \\ & \lambda_i \geq 0, \quad \forall i \in I_{ge} \\ & \lambda_i \leq 0, \quad \forall i \in I_{le} \\ & \lambda_i \text{ free}, \quad \forall i \in I_{eq} \\ & \lambda^T \mathbf{A}_j \leq c_j, \quad \forall j \in J_p \\ & \lambda^T \mathbf{A}_j \geq c_j, \quad \forall j \in J_n \\ & \lambda^T \mathbf{A}_j = c_j, \quad \forall j \in J_f \\ \text{variables} & \lambda \in \mathbb{R}^m.\end{array}$$

If primal is a minimization, we get the dual as a maximization.

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Primal-Dual Pair of Problems

Primal ($\mathcal{P}$)		Dual ($\mathcal{D}$)	
minimize	$\underset{x}{c}^T x$	maximize	$\lambda^T b$
$(\lambda_i \rightarrow)$	$a_i^T x \geq b_i, \quad \forall i \in I_{ge}$	$\lambda_i \geq 0,$	$\forall i \in I_{ge}$
$(\lambda_i \rightarrow)$	$a_i^T x \leq b_i, \quad \forall i \in I_{le}$	$\lambda_i \leq 0,$	$\forall i \in I_{le}$
$(\lambda_i \rightarrow)$	$a_i^T x = b_i, \quad \forall i \in I_{eq}$	λ_i free,	$\forall i \in I_{eq}$
	$x_j \geq 0, \quad \forall j \in J_p$	$\lambda^T A_j \leq c_j, \quad \forall j \in J_p$	
	$x_j \leq 0, \quad \forall j \in J_n$	$\lambda^T A_j \geq c_j, \quad \forall j \in J_n$	
	x_j free, $\forall j \in J_f$	$\lambda^T A_j = c_j, \quad \forall j \in J_f$	
variables	$x \in \mathbb{R}^n$	variables	$\lambda \in \mathbb{R}^m.$

If primal is a minimization, we get the dual as a maximization.

What if I rewrite the dual problem as a minimization problem, and construct its dual?

*What if instead of constructing a **lower** bound for a **minimization**, I want to get an **upper bound** for a **maximization**?*

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 \text{Primal } (\mathcal{P}) & \\
 \text{minimize}_{\mathbf{x}} & \mathbf{c}^\top \mathbf{x} \\
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 (\lambda_i \rightarrow) & \mathbf{a}_i^\top \mathbf{x} = b_i, \quad \forall i \in I_{eq} \\
 & x_j \geq 0, \quad \forall j \in J_p \\
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 & x_j \text{ free}, \quad \forall j \in J_f \\
 \text{variables} & \mathbf{x} \in \mathbb{R}^n
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Exercise

Rewrite the dual problem as a minimization problem and construct its dual.

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 \text{Primal } (\mathcal{P}) & \\
 \text{minimize}_{\mathbf{x}} & \mathbf{c}^\top \mathbf{x} \\
 (\lambda_i \rightarrow) & \mathbf{a}_i^\top \mathbf{x} \geq b_i, \quad \forall i \in I_{ge} \\
 (\lambda_i \rightarrow) & \mathbf{a}_i^\top \mathbf{x} \leq b_i, \quad \forall i \in I_{le} \\
 (\lambda_i \rightarrow) & \mathbf{a}_i^\top \mathbf{x} = b_i, \quad \forall i \in I_{eq} \\
 & x_j \geq 0, \quad \forall j \in J_p \\
 & x_j \leq 0, \quad \forall j \in J_n \\
 & x_j \text{ free}, \quad \forall j \in J_f \\
 \text{variables} & \mathbf{x} \in \mathbb{R}^n
 \end{array}$$

$$\begin{array}{ll}
 \text{Dual } (\mathcal{D}) & \\
 \text{maximize}_{\boldsymbol{\lambda}} & \boldsymbol{\lambda}^\top \mathbf{b} \\
 & \lambda_i \geq 0, \quad \forall i \in I_{ge} \\
 & \lambda_i \leq 0, \quad \forall i \in I_{le} \\
 & \lambda_i \text{ free}, \quad \forall i \in I_{eq} \\
 & \boldsymbol{\lambda}^\top \mathbf{A}_j \leq c_j, \quad \forall j \in J_p \\
 & \boldsymbol{\lambda}^\top \mathbf{A}_j \geq c_j, \quad \forall j \in J_n \\
 & \boldsymbol{\lambda}^\top \mathbf{A}_j = c_j, \quad \forall j \in J_f \\
 \text{variables} & \boldsymbol{\lambda} \in \mathbb{R}^m.
 \end{array}$$

Exercise

Rewrite the dual problem as a minimization problem and construct its dual.

Symmetry in LP Dualization

Primal-Dual Pair of Problems

minimize $c^T x$

$(\lambda_i \rightarrow) a_i^T x \geq b_i, \quad \forall i \in I_{ge}$

$(\lambda_i \rightarrow) a_i^T x \leq b_i, \quad \forall i \in I_{le}$

$(\lambda_i \rightarrow) a_i^T x = b_i, \quad \forall i \in I_{eq}$

$x_j \geq 0, \quad \forall j \in J_p$

$x_j \leq 0, \quad \forall j \in J_n$

x_j free, $\forall j \in J_f$

variables $x \in \mathbb{R}^n$

maximize $\lambda^T b$

$\lambda_i \geq 0, \quad \forall i \in I_{ge}$

$\lambda_i \leq 0, \quad \forall i \in I_{le}$

λ_i free, $\forall i \in I_{eq}$

$(x_j \rightarrow) \lambda^T A_j \leq c_j, \quad \forall j \in J_p$

$(x_j \rightarrow) \lambda^T A_j \geq c_j, \quad \forall j \in J_n$

$(x_j \rightarrow) \lambda^T A_j = c_j, \quad \forall j \in J_f$

variables $\lambda \in \mathbb{R}^m$.

For LPs, dualization in a **very symmetric** construction

Symmetry in LP Dualization

Primal-Dual Pair of Problems

minimize $c^T x$	maximize $\lambda^T b$
$(\lambda_i \rightarrow) a_i^T x \geq b_i, \quad \forall i \in I_{ge}$	$\lambda_i \geq 0, \quad \forall i \in I_{ge}$
$(\lambda_i \rightarrow) a_i^T x \leq b_i, \quad \forall i \in I_{le}$	$\lambda_i \leq 0, \quad \forall i \in I_{le}$
$(\lambda_i \rightarrow) a_i^T x = b_i, \quad \forall i \in I_{eq}$	λ_i free, $\forall i \in I_{eq}$
$x_j \geq 0, \quad \forall j \in J_p$	$(x_j \rightarrow) \lambda^T A_j \leq c_j, \quad \forall j \in J_p$
$x_j \leq 0, \quad \forall j \in J_n$	$(x_j \rightarrow) \lambda^T A_j \geq c_j, \quad \forall j \in J_n$
x_j free, $\forall j \in J_f$	$(x_j \rightarrow) \lambda^T A_j = c_j, \quad \forall j \in J_f$
variables $x \in \mathbb{R}^n$	variables $\lambda \in \mathbb{R}^m$

For LPs, dualization in a **very symmetric** construction

- Start with **any** primal LP : minimization/maximization

Symmetry in LP Dualization

Primal-Dual Pair of Problems

minimize	$\mathbf{c}^T \mathbf{x}$	maximize	$\boldsymbol{\lambda}^T \mathbf{b}$
$(\lambda_i \rightarrow)$	$\mathbf{a}_i^T \mathbf{x} \geq b_i, \quad \forall i \in I_{ge}$	$\lambda_i \geq 0,$	$\forall i \in I_{ge}$
$(\lambda_i \rightarrow)$	$\mathbf{a}_i^T \mathbf{x} \leq b_i, \quad \forall i \in I_{le}$	$\lambda_i \leq 0,$	$\forall i \in I_{le}$
$(\lambda_i \rightarrow)$	$\mathbf{a}_i^T \mathbf{x} = b_i, \quad \forall i \in I_{eq}$	λ_i free,	$\forall i \in I_{eq}$
	$x_j \geq 0, \quad \forall j \in J_p$	$(x_j \rightarrow)$	$\boldsymbol{\lambda}^T \mathbf{A}_j \leq c_j, \quad \forall j \in J_p$
	$x_j \leq 0, \quad \forall j \in J_n$	$(x_j \rightarrow)$	$\boldsymbol{\lambda}^T \mathbf{A}_j \geq c_j, \quad \forall j \in J_n$
	x_j free, $\forall j \in J_f$	$(x_j \rightarrow)$	$\boldsymbol{\lambda}^T \mathbf{A}_j = c_j, \quad \forall j \in J_f$
variables	$\mathbf{x} \in \mathbb{R}^n$	variables	$\boldsymbol{\lambda} \in \mathbb{R}^m.$

For LPs, dualization in a **very symmetric** construction

- Start with **any** primal LP : minimization/maximization
- For every **constraint** in the primal, associate a **decision variable** in the dual

Symmetry in LP Dualization

Primal-Dual Pair of Problems

minimize	$\mathbf{c}^T \mathbf{x}$		maximize	$\boldsymbol{\lambda}^T \mathbf{b}$	
$(\lambda_i \rightarrow)$	$\mathbf{a}_i^T \mathbf{x} \geq b_i,$	$\forall i \in I_{ge}$		$\lambda_i \geq 0,$	$\forall i \in I_{ge}$
$(\lambda_i \rightarrow)$	$\mathbf{a}_i^T \mathbf{x} \leq b_i,$	$\forall i \in I_{le}$		$\lambda_i \leq 0,$	$\forall i \in I_{le}$
$(\lambda_i \rightarrow)$	$\mathbf{a}_i^T \mathbf{x} = b_i,$	$\forall i \in I_{eq}$		λ_i free,	$\forall i \in I_{eq}$
	$x_j \geq 0,$	$\forall j \in J_p$	$(x_j \rightarrow)$	$\boldsymbol{\lambda}^T \mathbf{A}_j \leq c_j,$	$\forall j \in J_p$
	$x_j \leq 0,$	$\forall j \in J_n$	$(x_j \rightarrow)$	$\boldsymbol{\lambda}^T \mathbf{A}_j \geq c_j,$	$\forall j \in J_n$
	x_j free,	$\forall j \in J_f$	$(x_j \rightarrow)$	$\boldsymbol{\lambda}^T \mathbf{A}_j = c_j,$	$\forall j \in J_f$
variables	$\mathbf{x} \in \mathbb{R}^n$		variables	$\boldsymbol{\lambda} \in \mathbb{R}^m.$	

For LPs, dualization in a **very symmetric** construction

- Start with **any** primal LP : minimization/maximization
- For every **constraint** in the primal, associate a **decision variable** in the dual
- Dual is the opposite kind of problem : maximization/minimization

Symmetry in LP Dualization

Primal-Dual Pair of Problems

minimize	$\overset{x}{c}^T x$		maximize	$\overset{\lambda}{\lambda}^T b$	
$(\lambda_i \rightarrow)$	$a_i^T x \geq b_i,$	$\forall i \in I_{ge}$		$\lambda_i \geq 0,$	$\forall i \in I_{ge}$
$(\lambda_i \rightarrow)$	$a_i^T x \leq b_i,$	$\forall i \in I_{le}$		$\lambda_i \leq 0,$	$\forall i \in I_{le}$
$(\lambda_i \rightarrow)$	$a_i^T x = b_i,$	$\forall i \in I_{eq}$		λ_i free,	$\forall i \in I_{eq}$
	$x_j \geq 0,$	$\forall j \in J_p$	$(x_j \rightarrow)$	$\lambda^T A_j \leq c_j,$	$\forall j \in J_p$
	$x_j \leq 0,$	$\forall j \in J_n$	$(x_j \rightarrow)$	$\lambda^T A_j \geq c_j,$	$\forall j \in J_n$
	x_j free,	$\forall j \in J_f$	$(x_j \rightarrow)$	$\lambda^T A_j = c_j,$	$\forall j \in J_f$
variables	$x \in \mathbb{R}^n$		variables	$\lambda \in \mathbb{R}^m.$	

For LPs, dualization in a **very symmetric** construction

- Start with **any** primal LP : minimization/maximization
- For every **constraint** in the primal, associate a **decision variable** in the dual
- Dual is the opposite kind of problem : maximization/minimization
- Dual objective: inner product of **primal r.h.s.** and **dual decisions**

Symmetry in LP Dualization

Primal-Dual Pair of Problems

minimize	$c^T x$		maximize	$\lambda^T b$	
$(\lambda_i \rightarrow)$	$a_i^T x \geq b_i,$	$\forall i \in I_{ge}$		$\lambda_i \geq 0,$	$\forall i \in I_{ge}$
$(\lambda_i \rightarrow)$	$a_i^T x \leq b_i,$	$\forall i \in I_{le}$		$\lambda_i \leq 0,$	$\forall i \in I_{le}$
$(\lambda_i \rightarrow)$	$a_i^T x = b_i,$	$\forall i \in I_{eq}$		λ_i free,	$\forall i \in I_{eq}$
	$x_j \geq 0,$	$\forall j \in J_p$	$(x_j \rightarrow)$	$\lambda^T A_j \leq c_j,$	$\forall j \in J_p$
	$x_j \leq 0,$	$\forall j \in J_n$	$(x_j \rightarrow)$	$\lambda^T A_j \geq c_j,$	$\forall j \in J_n$
	x_j free,	$\forall j \in J_f$	$(x_j \rightarrow)$	$\lambda^T A_j = c_j,$	$\forall j \in J_f$
variables	$x \in \mathbb{R}^n$		variables	$\lambda \in \mathbb{R}^m.$	

For LPs, dualization in a **very symmetric** construction

- Start with **any** primal LP : minimization/maximization
- For every **constraint** in the primal, associate a **decision variable** in the dual
- Dual is the opposite kind of problem : maximization/minimization
- Dual objective: inner product of **primal r.h.s.** and **dual decisions**
- Depending on **primal constraint type** ($\geq, \leq, =$), add constraint on **sign of dual decision**

Symmetry in LP Dualization

Primal-Dual Pair of Problems

minimize	$c^T x$		maximize	$\lambda^T b$	
$(\lambda_i \rightarrow)$	$a_i^T x \geq b_i,$	$\forall i \in I_{ge}$		$\lambda_i \geq 0,$	$\forall i \in I_{ge}$
$(\lambda_i \rightarrow)$	$a_i^T x \leq b_i,$	$\forall i \in I_{le}$		$\lambda_i \leq 0,$	$\forall i \in I_{le}$
$(\lambda_i \rightarrow)$	$a_i^T x = b_i,$	$\forall i \in I_{eq}$		λ_i free,	$\forall i \in I_{eq}$
	$x_j \geq 0,$	$\forall j \in J_p$	$(x_j \rightarrow)$	$\lambda^T A_j \leq c_j,$	$\forall j \in J_p$
	$x_j \leq 0,$	$\forall j \in J_n$	$(x_j \rightarrow)$	$\lambda^T A_j \geq c_j,$	$\forall j \in J_n$
	x_j free,	$\forall j \in J_f$	$(x_j \rightarrow)$	$\lambda^T A_j = c_j,$	$\forall j \in J_f$
variables	$x \in \mathbb{R}^n$		variables	$\lambda \in \mathbb{R}^m.$	

For LPs, dualization in a **very symmetric** construction

- Start with **any** primal LP : minimization/maximization
- For every **constraint** in the primal, associate a **decision variable** in the dual
- Dual is the opposite kind of problem : maximization/minimization
- Dual objective: inner product of **primal r.h.s.** and **dual decisions**
- Depending on **primal constraint type** ($\geq, \leq, =$), add constraint on **sign of dual decision**
- Depending on **sign of primal decision**, set up **constraint on dual decisions**

Mnemonic Rules for Constructing **Any** Dual

$$\begin{array}{rcl}
 \underset{x}{\text{minimize}} & / & \underset{x}{\text{maximize}} \quad c^T x \\
 & & (\lambda \rightarrow) \quad Ax \begin{array}{l} \leq \\ \geq \\ \leq \\ \geq \\ \leq \\ \geq \end{array} b \\
 & & \quad \quad \quad x \begin{array}{l} \leq \\ \geq \\ \leq \\ \geq \\ \leq \\ \geq \end{array} 0
 \end{array} \tag{7}$$

Mnemonic Rules for Constructing **Any** Dual

$$\begin{array}{rcl}
 \underset{x}{\text{minimize}} & / & \underset{x}{\text{maximize}} \quad c^T x \\
 (\lambda \rightarrow) & & Ax \begin{array}{l} \leq \\ \geq \\ \leq \\ \geq \\ \leq \\ \geq \end{array} b \\
 & & x \begin{array}{l} \geq \\ \leq \\ \geq \\ \leq \\ \geq \\ \leq \end{array} 0
 \end{array} \tag{7}$$

R1: Add a **dual decision** λ_i for every primal constraint, i.e., every row a_i^T of A .

The dual objective will be to **maximize** $\lambda^T b$ / **minimize** $\lambda^T b$

Mnemonic Rules for Constructing **Any** Dual

$$\begin{array}{rcl}
 \underset{x}{\text{minimize}} & / & \underset{x}{\text{maximize}} \quad c^T x \\
 (\lambda \rightarrow) & & Ax \begin{array}{l} \leq \\ \geq \\ = \end{array} b \\
 & & \underset{x}{x} \begin{array}{l} \leq \\ \geq \\ = \end{array} 0
 \end{array} \tag{7}$$

R1: Add a **dual decision** λ_i for every primal constraint, i.e., every row a_i^T of A .

The dual objective will be to **maximize** $\lambda^T b$ / **minimize** $\lambda^T b$

R2: Signs: λ_i **free** for **equality** constraints ($a_i^T x = b_i$); otherwise, $\lambda_i \text{ } \textcircled{?} 0$.

Mnemonic Rules for Constructing **Any** Dual

$$\begin{array}{l} \underset{x}{\text{minimize}} \quad / \quad \underset{x}{\text{maximize}} \quad c^T x \\ (\lambda \rightarrow) \quad Ax \begin{array}{l} \leq \\ \geq \\ = \end{array} b \\ \quad \quad \quad x \begin{array}{l} \geq \\ \leq \\ = \end{array} 0 \end{array} \quad (7)$$

R1: Add a **dual decision** λ_i for every primal constraint, i.e., every row a_i^T of A .

The dual objective will be to **maximize** $\lambda^T b$ / **minimize** $\lambda^T b$

R2: Signs: λ_i **free** for **equality** constraints ($a_i^T x = b_i$); otherwise, λ_i $\textcircled{?}$ 0.

To determine the signs $\textcircled{?}$, ask yourself this Q:

*What happens to the **optimal objective value** if I increase the constraint r.h.s. b_i ?*

Powerful rule of thumb, which we will prove later:

λ_i is the slope of the optimal objective value w.r.t. the right-hand-side b_i

Depends on two things: is primal a **minimization**/maximization? Is constraint $\leq$ or $\geq$?

Mnemonic Rules for Constructing **Any** Dual

$$\begin{array}{l} \underset{x}{\text{minimize}} \quad / \quad \underset{x}{\text{maximize}} \quad c^T x \\ (\lambda \rightarrow) \quad Ax \begin{array}{l} \leq \\ \geq \\ = \end{array} b \\ \quad \quad \quad \underset{x}{\begin{array}{l} \leq \\ \geq \\ = \end{array}} 0 \end{array} \quad (7)$$

R1: Add a **dual decision** λ_i for every primal constraint, i.e., every row a_i^T of A .
The dual objective will be to **maximize** $\lambda^T b$ / **minimize** $\lambda^T b$

R2: Signs: λ_i **free** for **equality** constraints ($a_i^T x = b_i$); otherwise, $\lambda_i \text{ (?) } 0$.

To determine the signs (?) , ask yourself this Q:

*What happens to the **optimal objective value** if I increase the constraint r.h.s. b_i ?*

Powerful rule of thumb, which we will prove later:

λ_i is the slope of the optimal objective value w.r.t. the right-hand-side b_i

Depends on two things: is primal a **minimization**/maximization? Is constraint $\leq$ or $\geq$?

– primal minimization, “ $\leq$ ” constraint:

Mnemonic Rules for Constructing **Any** Dual

$$\begin{array}{l} \underset{x}{\text{minimize}} \quad / \quad \underset{x}{\text{maximize}} \quad c^T x \\ (\lambda \rightarrow) \quad Ax \begin{array}{l} \leq \\ \geq \\ = \end{array} b \\ \quad \quad \quad x \begin{array}{l} \geq \\ \leq \\ = \end{array} 0 \end{array} \quad (7)$$

R1: Add a **dual decision** λ_i for every primal constraint, i.e., every row a_i^T of A .

The dual objective will be to **maximize** $\lambda^T b$ / **minimize** $\lambda^T b$

R2: Signs: λ_i **free** for **equality** constraints ($a_i^T x = b_i$); otherwise, $\lambda_i \geq 0$.

To determine the signs $\geq$, ask yourself this Q:

*What happens to the **optimal objective value** if I increase the constraint r.h.s. b_i ?*

Powerful rule of thumb, which we will prove later:

λ_i is the slope of the optimal objective value w.r.t. the right-hand-side b_i

Depends on two things: is primal a **minimization**/maximization? Is constraint $\leq$ or $\geq$?

– primal minimization, “ $\leq$ ” constraint: objective would decrease

Mnemonic Rules for Constructing **Any** Dual

$$\begin{array}{l} \underset{x}{\text{minimize}} \quad / \quad \underset{x}{\text{maximize}} \quad c^T x \\ (\lambda \rightarrow) \quad Ax \begin{array}{l} \leq \\ \geq \\ = \end{array} b \\ \quad \quad \quad x \begin{array}{l} \geq \\ \leq \\ = \end{array} 0 \end{array} \quad (7)$$

R1: Add a **dual decision** λ_i for every primal constraint, i.e., every row a_i^T of A .
The dual objective will be to **maximize** $\lambda^T b$ / **minimize** $\lambda^T b$

R2: Signs: λ_i **free** for **equality** constraints ($a_i^T x = b_i$); otherwise, $\lambda_i \geq 0$.

To determine the signs $\geq$, ask yourself this Q:

*What happens to the **optimal objective value** if I increase the constraint r.h.s. b_i ?*

Powerful rule of thumb, which we will prove later:

λ_i is the slope of the optimal objective value w.r.t. the right-hand-side b_i

Depends on two things: is primal a **minimization**/maximization? Is constraint $\leq$ or $\geq$?

- primal minimization, " $\leq$ " constraint: objective would decrease $\rightarrow \lambda_i \leq 0$
- primal minimization, " $\geq$ " constraint:

Mnemonic Rules for Constructing **Any** Dual

$$\begin{array}{l} \underset{x}{\text{minimize}} \quad / \quad \underset{x}{\text{maximize}} \quad c^T x \\ (\lambda \rightarrow) \quad Ax \begin{array}{l} \leq \\ \geq \\ = \end{array} b \\ \quad \quad \quad x \begin{array}{l} \geq \\ \leq \\ = \end{array} 0 \end{array} \quad (7)$$

R1: Add a **dual decision** λ_i for every primal constraint, i.e., every row a_i^T of A .
The dual objective will be to **maximize** $\lambda^T b$ / **minimize** $\lambda^T b$

R2: Signs: λ_i **free** for **equality** constraints ($a_i^T x = b_i$); otherwise, λ_i $\textcircled{?}$ 0.

To determine the signs $\textcircled{?}$, ask yourself this Q:

*What happens to the **optimal objective value** if I increase the constraint r.h.s. b_i ?*

Powerful rule of thumb, which we will prove later:

λ_i is the slope of the optimal objective value w.r.t. the right-hand-side b_i

Depends on two things: is primal a **minimization**/maximization? Is constraint $\leq$ or $\geq$?

- primal minimization, " $\leq$ " constraint: objective would decrease $\rightarrow \lambda_i \leq 0$
- primal minimization, " $\geq$ " constraint: objective would increase

Mnemonic Rules for Constructing **Any** Dual

$$\begin{array}{l} \underset{x}{\text{minimize}} \quad / \quad \underset{x}{\text{maximize}} \quad c^T x \\ (\lambda \rightarrow) \quad Ax \begin{array}{l} \leq \\ \geq \\ = \end{array} b \\ \quad \quad \quad x \begin{array}{l} \geq \\ \leq \\ = \end{array} 0 \end{array} \quad (7)$$

R1: Add a **dual decision** λ_i for every primal constraint, i.e., every row a_i^T of A .
The dual objective will be to **maximize** $\lambda^T b$ / **minimize** $\lambda^T b$

R2: Signs: λ_i **free** for **equality** constraints ($a_i^T x = b_i$); otherwise, $\lambda_i \text{ (?) } 0$.

To determine the signs $(?)$, ask yourself this Q:

*What happens to the **optimal objective value** if I **increase** the constraint r.h.s. b_i ?*

Powerful rule of thumb, which we will prove later:

λ_i is the slope of the optimal objective value w.r.t. the right-hand-side b_i

Depends on two things: is primal a **minimization**/maximization? Is constraint $\leq$ or $\geq$?

- primal minimization, " $\leq$ " constraint: objective would decrease $\rightarrow \lambda_i \leq 0$
- primal minimization, " $\geq$ " constraint: objective would increase $\rightarrow \lambda_i \geq 0$
- primal maximization, " $\leq$ " constraint:

Mnemonic Rules for Constructing **Any** Dual

$$\begin{array}{l} \underset{x}{\text{minimize}} \quad / \quad \underset{x}{\text{maximize}} \quad c^T x \\ (\lambda \rightarrow) \quad Ax \begin{array}{l} \leq \\ \geq \\ = \end{array} b \\ \quad \quad \quad x \begin{array}{l} \geq \\ \leq \\ = \end{array} 0 \end{array} \quad (7)$$

R1: Add a **dual decision** λ_i for every primal constraint, i.e., every row a_i^T of A .
The dual objective will be to **maximize** $\lambda^T b$ / **minimize** $\lambda^T b$

R2: Signs: λ_i **free** for **equality** constraints ($a_i^T x = b_i$); otherwise, $\lambda_i \text{ (?) } 0$.

To determine the signs (?), ask yourself this Q:

*What happens to the **optimal objective value** if I increase the constraint r.h.s. b_i ?*

Powerful rule of thumb, which we will prove later:

λ_i is the slope of the optimal objective value w.r.t. the right-hand-side b_i

Depends on two things: is primal a **minimization**/maximization? Is constraint $\leq$ or $\geq$?

- primal minimization, " $\leq$ " constraint: objective would decrease $\rightarrow \lambda_i \leq 0$
- primal minimization, " $\geq$ " constraint: objective would increase $\rightarrow \lambda_i \geq 0$
- primal maximization, " $\leq$ " constraint: objective would increase

Mnemonic Rules for Constructing **Any** Dual

$$\begin{array}{l} \underset{x}{\text{minimize}} \quad / \quad \underset{x}{\text{maximize}} \quad c^T x \\ (\lambda \rightarrow) \quad Ax \begin{array}{l} \leq \\ \geq \\ = \end{array} b \\ \quad \quad \quad x \begin{array}{l} \geq \\ \leq \\ = \end{array} 0 \end{array} \quad (7)$$

R1: Add a **dual decision** λ_i for every primal constraint, i.e., every row a_i^T of A .
The dual objective will be to **maximize** $\lambda^T b$ / **minimize** $\lambda^T b$

R2: Signs: λ_i **free** for **equality** constraints ($a_i^T x = b_i$); otherwise, $\lambda_i \geq 0$.

To determine the signs $\geq$, ask yourself this Q:

*What happens to the **optimal objective value** if I increase the constraint r.h.s. b_i ?*

Powerful rule of thumb, which we will prove later:

λ_i is the slope of the optimal objective value w.r.t. the right-hand-side b_i

Depends on two things: is primal a **minimization**/maximization? Is constraint $\leq$ or $\geq$?

- primal minimization, " $\leq$ " constraint: objective would decrease $\rightarrow \lambda_i \leq 0$
- primal minimization, " $\geq$ " constraint: objective would increase $\rightarrow \lambda_i \geq 0$
- primal maximization, " $\leq$ " constraint: objective would increase $\rightarrow \lambda_i \geq 0$
- primal maximization, " $\geq$ " constraint:

Mnemonic Rules for Constructing **Any** Dual

$$\begin{array}{l} \underset{x}{\text{minimize}} \quad / \quad \underset{x}{\text{maximize}} \quad c^T x \\ (\lambda \rightarrow) \quad Ax \begin{array}{l} \leq \\ \geq \\ = \end{array} b \\ \quad \quad \quad x \begin{array}{l} \geq \\ \leq \\ = \end{array} 0 \end{array} \quad (7)$$

R1: Add a **dual decision** λ_i for every primal constraint, i.e., every row a_i^T of A .
The dual objective will be to **maximize** $\lambda^T b$ / **minimize** $\lambda^T b$

R2: Signs: λ_i **free** for **equality** constraints ($a_i^T x = b_i$); otherwise, $\lambda_i \geq 0$.

To determine the signs $\geq$, ask yourself this Q:

*What happens to the **optimal objective value** if I increase the constraint r.h.s. b_i ?*

Powerful rule of thumb, which we will prove later:

λ_i is the slope of the optimal objective value w.r.t. the right-hand-side b_i

Depends on two things: is primal a **minimization**/maximization? Is constraint $\leq$ or $\geq$?

- primal minimization, " $\leq$ " constraint: objective would decrease $\rightarrow \lambda_i \leq 0$
- primal minimization, " $\geq$ " constraint: objective would increase $\rightarrow \lambda_i \geq 0$
- primal maximization, " $\leq$ " constraint: objective would increase $\rightarrow \lambda_i \geq 0$
- primal maximization, " $\geq$ " constraint: objective would decrease

Mnemonic Rules for Constructing **Any** Dual

$$\begin{array}{l} \underset{x}{\text{minimize}} \quad / \quad \underset{x}{\text{maximize}} \quad c^T x \\ (\lambda \rightarrow) \quad Ax \begin{array}{l} \leq \\ \geq \\ = \end{array} b \\ \quad \quad \quad x \begin{array}{l} \geq \\ \leq \\ = \end{array} 0 \end{array} \quad (8)$$

R1: Add a **dual decision** λ_i for every primal constraint, i.e., every row a_i^T of A .
The dual objective will be to **maximize** $\lambda^T b$ / **minimize** $\lambda^T b$

R2: Signs: λ_i **free** for **equality** constraints ($a_i^T x = b_i$); otherwise, $\lambda_i \text{ (?) } 0$.

- λ_i is the slope of the optimal objective value w.r.t. the right-hand-side b_i
- primal minimization, " $\leq$ " constraint: objective would decrease $\rightarrow \lambda_i \leq 0$
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R3: In the dual, add a **constraint** for every primal variable x_j
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Mnemonic Rules for Constructing **Any** Dual

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Determine (?) using rules in [R2]: treat dual as primal, c_j as r.h.s., x_j as dual variable

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Mnemonic Rules for Constructing **Any** Dual

$$\begin{array}{c} \text{minimize} \\ \mathbf{x} \end{array} / \begin{array}{c} \text{maximize} \\ \mathbf{x} \end{array} \quad c^T \mathbf{x} \quad (8)$$

$$(\lambda \rightarrow) \quad A \mathbf{x} \begin{array}{c} \leq \\ \leq \\ \leq \\ \leq \\ \leq \\ \leq \\ \leq \\ \leq \\ \leq \\ \leq \end{array} b$$

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- dual maximization: $x_j \geq 0 \Rightarrow \lambda^T A_j \leq c_j$. $x_j \leq 0 \Rightarrow \lambda^T A_j \leq c_j$
- dual maximization: $x_j \geq 0 \Rightarrow \lambda^T A_j \leq c_j$. $x_j \leq 0 \Rightarrow \lambda^T A_j \leq c_j$

$$(\mathcal{P}) \quad \min_{x_1, x_2} \quad c_1^T x_1 + c_2^T x_2$$

$$\text{s.t.} \quad A_{11}x_1 + A_{12}x_2 \geq b_1,$$

$$A_{21}x_1 + A_{22}x_2 = b_2,$$

$$x_1 \geq 0,$$

$$x_2 \text{ free.}$$

Some Quick Results

Theorem ("Duals of equivalent primals")

If we transform a primal P_1 into an equivalent formulation P_2 by:

- *replacing a free variable x_i with $x_i = x_i^+ - x_i^-$,*
- *replacing an inequality with an equality by introducing a slack variable,*
- *removing linearly dependent rows a_i^T for a **feasible** LP in standard form,*

*then the duals of (P_1) and (P_2) are **equivalent**, i.e., they are either both infeasible or they have the same optimal objective.*

Weak duality

Primal ($\mathcal{P}$)

$$\begin{aligned} &\text{minimize}_{\mathbf{x}} && \mathbf{c}^T \mathbf{x} \\ &(\lambda_i \rightarrow) && \mathbf{a}_i^T \mathbf{x} \geq b_i, && \forall i \in I_{\text{ge}}, \\ &(\lambda_i \rightarrow) && \mathbf{a}_i^T \mathbf{x} \leq b_i, && \forall i \in I_{\text{le}}, \\ &(\lambda_i \rightarrow) && \mathbf{a}_i^T \mathbf{x} = b_i, && \forall i \in I_{\text{eq}}, \\ &&& \mathbf{x}_j \geq 0, && \forall j \in J_p, \\ &&& \mathbf{x}_j \leq 0, && \forall j \in J_n, \\ &&& \mathbf{x}_j \text{ free}, && \forall j \in J_f. \end{aligned}$$

Dual ($\mathcal{D}$)

$$\begin{aligned} &\text{maximize}_{\boldsymbol{\lambda}} && \boldsymbol{\lambda}^T \mathbf{b} \\ &&& \lambda_i \geq 0, && \forall i \in I_{\text{ge}}, \\ &&& \lambda_i \leq 0, && \forall i \in I_{\text{le}}, \\ &&& \lambda_i \text{ free}, && \forall i \in I_{\text{eq}}, \\ &(\mathbf{x}_j \rightarrow) && \boldsymbol{\lambda}^T \mathbf{A}_j \leq c_j, && \forall j \in J_p, \\ &(\mathbf{x}_j \rightarrow) && \boldsymbol{\lambda}^T \mathbf{A}_j \geq c_j, && \forall j \in J_n, \\ &(\mathbf{x}_j \rightarrow) && \boldsymbol{\lambda}^T \mathbf{A}_j = c_j, && \forall j \in J_f. \end{aligned}$$

Weak duality

Primal ($\mathcal{P}$)			Dual ($\mathcal{D}$)		
minimize _{x}	$c^T x$		maximize _{λ}	$\lambda^T b$	
$(\lambda_i \rightarrow)$	$a_i^T x \geq b_i,$	$\forall i \in I_{ge},$	$\lambda_i \geq 0,$		$\forall i \in I_{ge},$
$(\lambda_i \rightarrow)$	$a_i^T x \leq b_i,$	$\forall i \in I_{le},$	$\lambda_i \leq 0,$		$\forall i \in I_{le},$
$(\lambda_i \rightarrow)$	$a_i^T x = b_i,$	$\forall i \in I_{eq},$	λ_i free,		$\forall i \in I_{eq},$
	$x_j \geq 0,$	$\forall j \in J_p,$	$(x_j \rightarrow)$	$\lambda^T A_j \leq c_j,$	$\forall j \in J_p,$
	$x_j \leq 0,$	$\forall j \in J_n,$	$(x_j \rightarrow)$	$\lambda^T A_j \geq c_j,$	$\forall j \in J_n,$
	x_j free,	$\forall j \in J_f.$	$(x_j \rightarrow)$	$\lambda^T A_j = c_j,$	$\forall j \in J_f.$

Theorem (Weak duality)

If x is feasible for ($\mathcal{P}$) and λ is feasible for ($\mathcal{D}$), then $\lambda^T b \leq c^T x$.

Proof. Trivially true from our construction – omitted.

Implications of Weak Duality

Corollary

The following results hold:

- (a) If the optimal objective in $(\mathcal{P})$ is $-\infty$, then $(\mathcal{D})$...*
- (b) If the optimal objective in $(\mathcal{D})$ is $+\infty$, then $(\mathcal{P})$...*

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(c) If $x \in P$ and $\lambda \in D$, then:

$$c^T x - p^* \leq c^T x - \lambda^T b \quad \textbf{and} \quad d^* - \lambda^T b \leq c^T x - \lambda^T b.$$

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(d) *If $x \in P$, $\lambda \in D$, and $\lambda^T b = c^T x$, then x **optimal** for $(\mathcal{P})$ and λ **optimal** for $(\mathcal{D})$.*

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(c) and (d) provide (sub)optimality certificates, but...

How do we know that the gaps in (c) are not very large?

How do we know that x and λ satisfying (d) even exist?

Strong duality

Theorem (Strong duality)

If $(\mathcal{P})$ has an optimal solution, so does $(\mathcal{D})$, and the optimal values are equal, $\lambda^ = d^*$.*

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Proof. Many proofs possible...

- See Bertsimas & Tsitsiklis for a proof involving the simplex algorithm
- We provide a more general proof, in three steps:
 1. The **separating hyperplane theorem** (for convex sets)
 2. The Farkas Lemma
 3. Strong duality

Need a tiny bit of **real analysis** background...

A Few Real Analysis Results

Definition (Closed Set)

A set $S \subseteq \mathbb{R}^n$ is called **closed** if it contains the limit of any sequence of elements of S . That is, if $x_n \in S$, $\forall n \geq 1$ and $x_n \rightarrow x^*$, then $x^* \in S$.

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Every polyhedron is closed.

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Theorem

Every polyhedron is closed.

Proof.

- Consider $P = \{x \in \mathbb{R}^n \mid Ax \geq b\}$ (representation is w.l.o.g.)
- Suppose that $\{x_n\}_{n \geq 1}$ is a sequence with $x_n \in P$ for every n , and $x_n \rightarrow x^*$.
- For each k , we have $x_k \in P$, and therefore, $Ax_k \geq b$.
- Then, $Ax^* = A(\lim_{k \rightarrow \infty} x_k) = \lim_{k \rightarrow \infty} Ax_k \geq b$, so x^* belongs to P . □

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*Is every **convex set** closed?*

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A set $S \subseteq \mathbb{R}^n$ is called **closed** if it contains the limit of any sequence of elements of S . That is, if $x_n \in S$, $\forall n \geq 1$ and $x_n \rightarrow x^*$, then $x^* \in S$.

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Every polyhedron is closed.

Theorem (Weierstrass' Theorem)

If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a continuous function, and if S is a nonempty, closed, and bounded subset of $\mathbb{R}^n$, then there exist $\underline{x}, \bar{x} \in S$ such that $f(\underline{x}) \leq f(x) \leq f(\bar{x})$ for all $x \in S$.

i.e., a continuous function achieves its minimum and maximum

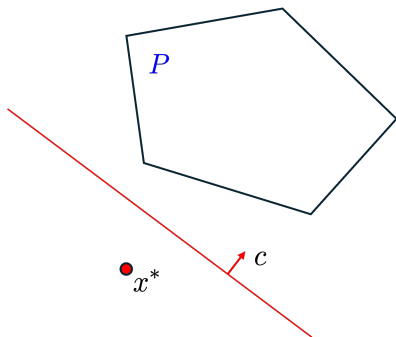
Separating Hyperplane Theorem

The first **fundamental result in optimization**

Separating Hyperplane Theorem

Theorem (**Simple** Separating Hyperplane Theorem)

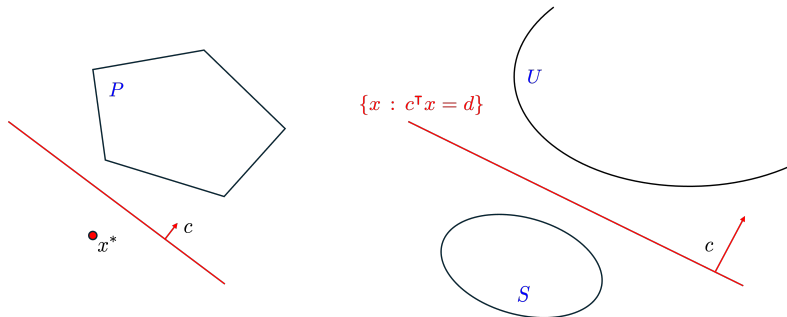
Consider a point x^* and a polyhedron P . If $x^* \notin P$, then there exists a vector $c \in \mathbb{R}^n$ such that $c \neq 0$ and $c^T x^* < c^T y$ holds for all $y \in P$.



Separating Hyperplane Theorem

Theorem (Separating Hyperplane Theorem for Convex Sets)

Let S and U be two nonempty, closed, convex subsets of $\mathbb{R}^n$ such that $S \cap U = \emptyset$ and S is bounded. Then, there exists $c \in \mathbb{R}^n$ and $d \in \mathbb{R}$ such that $S \subset \{x \in \mathbb{R}^n : c^T x < d\}$ and $U \subset \{x \in \mathbb{R}^n : c^T x > d\}$.



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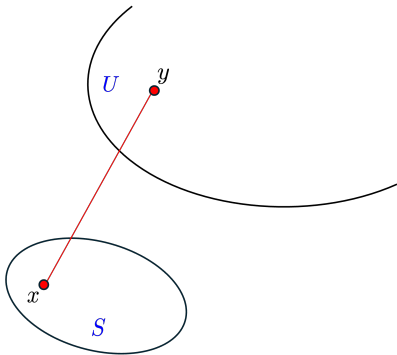
Proof.

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Proof. Consider $\|x - y\|$ with $x \in S, y \in U$

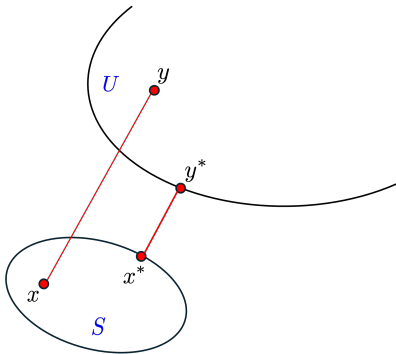


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Proof. Argue that the minimum is achieved, at x^*, y^*

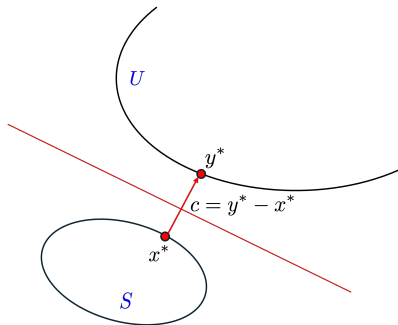


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Proof. Argue that $c = y^* - x^*$ and $d = \frac{c^T(x^* + y^*)}{2}$ give strict separating hyperplane



Separating Hyperplane Theorem - Caveats!

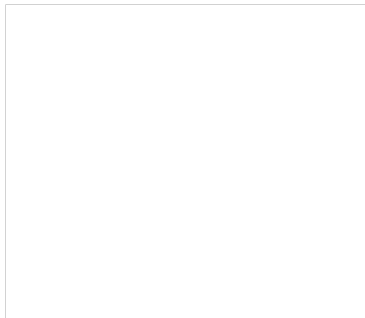
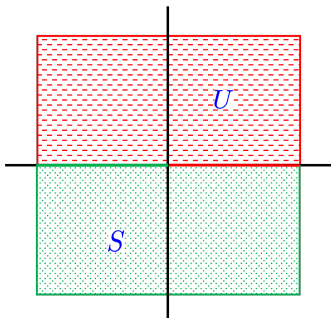
Both conditions in the theorem needed: **closed** and at least one set **bounded**

Separating Hyperplane Theorem - Caveats!

Both conditions in the theorem needed: **closed** and at least one set **bounded**

- **Left:** two convex sets that are **not closed** but are both bounded:

$$S = [-1, 1] \times [-1, 0) \cup \{(x, y) : x \in [-1, 0], y = 0\}, \quad U = [-1, 1]^2 \setminus S$$



Separating Hyperplane Theorem - Caveats!

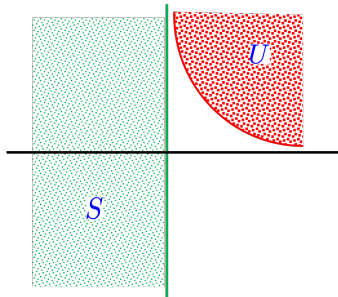
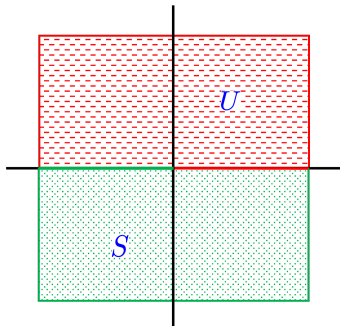
Both conditions in the theorem needed: **closed** and at least one set **bounded**

- **Left:** two convex sets that are **not closed** but are both bounded:

$$S = [-1, 1] \times [-1, 0) \cup \{(x, y) : x \in [-1, 0], y = 0\}, \quad U = [-1, 1]^2 \setminus S$$

- **Right:** two convex sets that are both closed but are **unbounded**

$$S = \{(x, y) : x \leq 0\}, \quad U = \{(x, y) : x \geq 0, y \geq 1/x\}$$

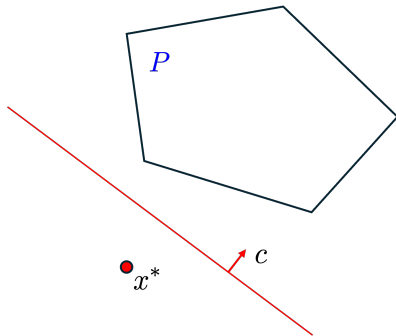


Needed For Our Purposes

We proved the first **fundamental result in optimization**!

Corollary (Needed for our purposes...)

If P is a polyhedron and $x^ \notin P$, there exists a hyperplane that strictly separates x^* from P , i.e., $\exists c \neq 0$ such that $c^T x^* < c^T x$ for any $x \in P$.*



Farkas Lemma

Time for the **second fundamental result in optimization!**

Farkas Lemma

Theorem (Farkas' Lemma)

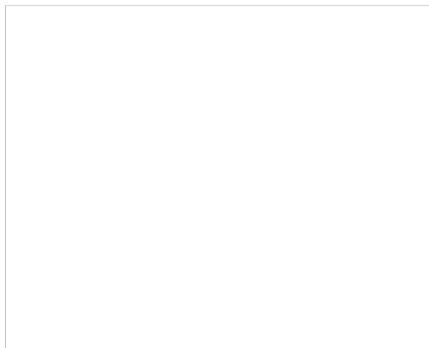
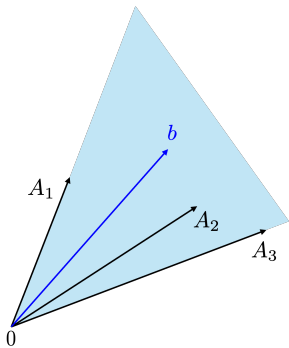
For $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, exactly one of the following two alternatives holds:

Farkas Lemma

Theorem (Farkas' Lemma)

For $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, exactly one of the following two alternatives holds:

(a) There exists some $x \geq 0$ such that $Ax = b$.



Farkas Lemma

Theorem (Farkas' Lemma)

For $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, exactly one of the following two alternatives holds:

- (a) There exists some $x \geq 0$ such that $Ax = b$.
- (b) There exists some vector λ such that $\lambda^T A \geq 0$ and $\lambda^T b < 0$.

